Demonstrative descriptions and anti-uniqueness[1]

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Abstract. Demonstrative descriptions robustly give rise to an anti-uniqueness inference, rendering them infelicitous with inherently unique nouns. Two major theoretical approaches have been proposed to explain this phenomenon. One posits that demonstratives lexically encode an anti-uniqueness presupposition in their semantics (Nowak, 2019; Dayal and Jiang, 2022; Owusu, 2022), while the other derives the inference from general pragmatic principles (Blumberg, 2020; Ahn, 2022, 2023). In this paper, I present evidence in favor of the presuppositional account and introduce a modalized implementation that addresses key limitations of prior proposals. Drawing on cross-linguistic empirical data, I further demonstrate that the anti-uniqueness approach is better suited to account for typological variation in this domain.

Keywords: complex demonstratives, definites, anti-uniqueness, presupposition, Bangla

1. Introduction

In the classical Kaplanian tradition, demonstratives were considered to be fundamentally distinct from definites. This view was based on their ability to refer directly and rigidly, unlike definites (Kaplan, 1989). In (1a), the referent of the definite description *'the butler'* is determined by identifying the unique entity that satisfies the description in the given context. In contrast, the demonstrative in (1b) simply denotes the referent directly by picking out the individual being pointed to in the utterance context.[2] Like indexicals, the denotation of the demonstrative is not sensitive to displacements in possible circumstances of evaluation encoded in the sentence—in (1b), the referent for '*that butler*' is not evaluated in worlds compatible with John's beliefs, but anchored to the individual pointed to in the world of utterance.

(1) a. John believes that [the butler] stole the jewelry.
b. John believes that [that butler]$_{\rightarrow}$ stole the jewelry.

While Kaplan's account focused on the deictic uses of demonstratives, later works, based on evidence from their non-deictic uses, take the stance that demonstratives are much more closely related to definites (Himmelmann, 1996; King, 2001; Roberts, 2002; Wolter, 2006; Elbourne, 2008). Demonstratives can have anaphoric (2a) and quantificational readings (2b) and can also compose with relative clauses that identify the intended referent (2c), analogous to definite descriptions (King, 2001; Roberts, 2002; Wolter, 2006; Simonenko, 2014; Nowak, 2019).

(2) a. I met a child. That child looked happy.

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[2]Adopting the convention in Ahn (2017) and subsequent works, the demonstrative gesture associated with a linguistic expression is indicated with the symbol $_{\rightarrow}$ following it.

b. Every dog in my neighborhood, even the meanest, has an owner who thinks that that dog is a sweetie. (Roberts, 2002: 5)

c. That student who scored one hundred on the exam is a genius. (King, 2001: 3)

The overlap between demonstratives and definites extends cross-linguistically as well. Similar to English, other languages with determiners, such as German, also exhibit distributional parallels between definite and demonstrative descriptions in their anaphoric uses (Schwarz, 2009) (see Saha and Davidson (2024) for nuances). In many determiner-less languages, too, demonstratives mark definiteness or anaphoricity (Ahn, 2017; Schwarz, 2009, 2013; Dayal and Jiang, 2022; Simpson and Wu, 2022). Furthermore, in some languages, demonstratives are the primary or preferred mode of anaphoric reference, e.g., Eastern Ojibwa (Nichols, 1988; Dryer, 2011), Mandarin (Jenks, 2015, 2018; Bremmers et al., 2022; Saha et al., 2024).[3] Many languages even use the same form, but in different positions, to express demonstrative and definite uses, e.g., Swahili, Ute, Shambala, and Pa'a (Dryer, 2011).

Naturally, given the considerable overlap in their use and the subtlety in their interpretive differences, a lot of recent attention has been on investigating the precise ways in which the two differ by identifying specific contexts that tease apart their distribution. Robinson (2005) discusses the contrast in (3a), where the definite is licensed, but the demonstrative is not. This contrast robustly extends to other predicates as well that, with respect to contexts drawn from our actual world and history, refer to singleton sets of entities, as shown in (3b)-(3c).

(3) a. The/ #That sun is made of hydrogen and helium. (Robinson, 2005: 51)

b. The/ #That author of Waverley also wrote Ivanhoe. (Nowak, 2019: 2867)

c. The/ #That current president of India is Droupadi Murmu.

The core claim is that demonstratives are not licensed when there is a unique entity that satisfies the denotation of its NP complement (Roberts, 2002; Robinson, 2005; Wolter, 2006; Nowak, 2019; Dayal and Jiang, 2022; Owusu, 2022). This has been accounted for in several recent works by proposing that demonstratives encode a presupposition of anti-uniqueness in their denotation, and accounts vary on the implementation of this presupposition. Nowak (2019) argues that demonstratives carry a second argument (in addition to the NP complement), which restricts the set denoted by the NP to its proper subset. Dayal and Jiang (2022) argue that there needs to be at least one other entity in the set denoted by the NP, in a situation larger than the situation in which uniqueness is evaluated. On the other hand, opposing analyses like Blumberg (2020), Ahn (2022, 2023) argue against an anti-uniqueness presupposition and propose accounts that derive the markedness of demonstratives from pragmatic principles.

[3] Although these forms are widely used in contexts where English prefers a definite determiner over a demonstrative, their classification as demonstratives is based on their ability to appear in deictic contexts accompanied by pointing gestures and their ability to encode spatial contrasts, much like English 'this' and 'that'. Another useful diagnostic to distinguish demonstratives from definites can be the contrastiveness test proposed in Dayal (prep). This test employs two incompatible predicates, i.e., properties that cannot apply to the same individual simultaneously, to tease apart the behavior of demonstratives from definites. For example:

(1) a. [This dog]$_{\rightarrow}$ is sleeping while [this dog]$_{\rightarrow}$ is running around.

b. #[The dog]$_{\rightarrow}$ is sleeping while [the dog]$_{\rightarrow}$ is running around. (Dayal, prep)

The outline of the rest of the paper is as follows: Section 2 reviews the two main approaches to anti-uniqueness and outlines their limitations: (i) presuppositional accounts, which posit a lexically specified anti-uniqueness condition in demonstratives, and (ii) pragmatic accounts, which derive it from general economy principles. Section 3 defends the core intuition behind anti-uniqueness based accounts and proposes a modal implementation that addresses the challenges raised in Section 2. Section 4 introduces further empirical complexity by presenting cross-linguistic data, which illustrate that anti-uniqueness is exhibited not only by deictic elements like demonstratives but also by a particular class of definites in Bangla, and argues that a presuppositional analysis better captures this typological variability. Section 5 concludes.

Note on terminology: For convenience, I refer to the relevant referential expressions as 'demonstratives', omitting the term 'descriptions'. However this should not be understood to mean demonstrative pronouns. This paper focuses exclusively on demonstratives that feature an overt noun phrase, i.e., demonstrative descriptions/ complex demonstratives.

2. Markedness of demonstratives: The two broad categories of approaches

2.1. Accounts based on an anti-uniqueness presupposition

The idea that demonstratives are not licensed when there is a unique entity that satisfies the denotation of its NP complement has been pursued in various forms in several works (Roberts, 2002; Robinson, 2005; Wolter, 2006; Nowak, 2019; Dayal and Jiang, 2022; Owusu, 2022). In this paper, I focus on presuppositional accounts of this restriction, in particular, the proposals in Nowak (2019) and Dayal and Jiang (2022). Nowak's (2019) theory is built to account for the semantic and structural constraints of demonstratives as seen in sentences such as (4a)-(4b).[4]

(4) a. The/ That guy who wrote Waverley also wrote Ivanhoe. (Nowak, 2019: 2867)
b. The/ #That author of Waverley also wrote Ivanhoe. (Nowak, 2019: 2867)

Building on King (2001), Nowak proposes that a demonstrative is a definite expression that carries an additional second argument (in addition to the NP complement) and presupposes that this second argument restricts the set denoted by the NP to its proper subset. This is shown in (6), compared with (5) for the definite.

(5) the F = the x: F(x)

(6) $\text{that F} = \begin{cases} \text{[the x: [F(x) \& G(x)]] iff } (F \cap G) \subset F \\ \text{otherwise undefined} \end{cases}$ (Nowak, 2019: 2880)

Nowak also proposes that the required G(x) property must syntactically occur outside the main DP headed by the demonstrative. For instance, in (4a), the relative clause '*who wrote Waverley*'

[4] As has been noted in Wolter (2006); Ahn (2017); Nowak (2019); Dayal (2023), a.o., there are certain emotive or affective uses of demonstratives that do not exhibit anti-uniqueness effects, as in (1) and (2). This paper does not focus on the emotive uses of demonstratives, since such uses raise separate issues that cannot be settled here. For discussion, see Lakoff (1974); Wolter (2006); Potts and Schwarz (2010).

(1) That Henry Kissinger is a really great guy! (Wolter, 2006: 82)

(2) That (darn) author of Waverley—he turns up everywhere in philosophy, doesn't he?! (Nowak, 2019: 2867, attributed to Partee (p.c.))

is argued to occupy the CP position, as in (7). It properly restricts the set denoted by the NP, and hence the felicity of (4a) is predicted.[5] In contrast, the genitive phrase '*of Waverley*' in (4b) occurs NP-internally and cannot occupy the CP position.

(7) $[_{DP}[_{DP} that\ guy][_{CP} who\ wrote\ Waverley]]$

Ahn (2022) argues that such a presuppositional account is too strong, based on evidence from anaphora, as in (8). The demonstrative description '*that woman*' is felicitous here even though there is only one woman that is relevant in the context.

(8) A woman came into the room. That woman sat down.

Such cases are accounted for in Dayal and Jiang (2022) by arguing that the anti-uniqueness constraint of demonstratives is not subject to satisfaction in the situation of evaluation, but in some larger situation. This is what drives the contrast in the felicity of (8) and (3a) for instance, which is repeated below in (9).

(9) The/ #That sun is made of hydrogen and helium. (Robinson, 2005: 51)

Appealing to the notion of domain widening (Kadmon and Landman, 1993), they argue that the reference situation in (8) can be considered to be part of situations that may contain more than one woman, even if no such woman is salient. In contrast, in our world, the reference situation in (9) cannot be considered to be part of situations than contain multiple suns. This is formalized as the anti-uniqueness presupposition in (10), where the free variable y (in line with the convention in Schwarz 2009) represents the object being picked (intended referent).

(10) $[\![\text{DEM}]\!] = \lambda s \lambda P : \exists s' s \leq s' |P_{s'}| > 1.\iota x[P_s(x) \wedge x = y]$ (Dayal & Jiang, 2022: 158)

(10) takes a reference situation s and a property P and returns the unique individual that satisfies P and is identical to y in s, if defined. The anti-uniqueness condition is encoded in the presupposition, which states that the set denoted by P must have cardinality greater than one in the wider situation s' that s is a part of ($\leq$).

2.1.1. Limitations of anti-uniqueness accounts

One of the limitations of accounts based on anti-uniqueness presuppositions, as Nowak himself points out, is seen in cases such as (11a) and (11b). Both these sentences exhibit a clearly felicitous use of the demonstrative, even though they appear to violate the presuppositions that there is only one rhino and only one atomic bomb left in the contexts specified by (11a) and (11b) respectively. Nowak's analysis does not provide a solution for this problem.

(11) a. That rhino is the only one still alive. (Nowak, 2019: 2897)
b. That atomic bomb is the last one remaining. (Nowak, 2019: 2897)

[5] Such a structure for restrictive relative clauses was considered standard in syntax earlier (Ross, 1967; Stockwell et al., 1973). But they were later ruled out on semantic grounds, as they violate compositionality (Partee, 1975). However, Bach and Cooper (1978) provided a solution to this problem by raising the type of the determiner via the insertion of a variable over properties into its semantic representation.

Dayal and Jiang's (2022) proposal of satisfying the anti-uniqueness condition in a larger situation does not help us either. Let's see why. The Kratzerian notion of situation has the following fundamental characteristics (Kratzer, 1989). Situations are considered to be particulars that exist in worlds and at times. A situation is part of a possible world, and any part of a possible world is a situation. Situations stand in a mereological part-whole relationships to each other, i.e., they can be temporal or spatial sub-parts of other situations. Any situation s is part of another situation s', if and only if $s + s' = s'$. In this way, situations can extend and grow into a particular possible world, which is the maximal element in a situational mereology. In other words, a world is a situation that is not a part of any other situation.

On Dayal and Jiang's account, the felicity of (11a) (and similarly for (11b)) would have to be explained by assuming that the presupposition of the demonstrative is met in some larger situation that contains more than one rhino. However, the sentence explicitly asserts that there is only one rhino in the maximal situation – the world. Hence, the felicity of such sentences pose a challenge for the situation semantic treatment of the anti-uniqueness presupposition. A potential resolution to this issue is discussed in footnote 9.

The second challenge for presuppositional approaches is raised by sentences like (12). Blumberg (2020) argues that sentences like (12) overtly reject the anti-uniqueness presupposition, and yet they are felicitous, contrary to the predictions of the presuppositional theories. (12) is contrasted with unquestionable cases of presupposition violations, as in (13).

(12) I don't know if there are any other cars in this showroom, but [that car]$_{\rightarrow}$ looks expensive. (Blumberg, 2020: 49)

(13) #I don't know if Mary ever smoked, but Mary stopped smoking. (Blumberg, 2020: 49)

Blumberg contends that "when a speaker expresses doubt that a certain condition holds and then utters a sentence that has that condition as a presupposition, infelicity results" (Blumberg 2020: 49). (13) is clearly infelicitous, on these grounds. However, (12) is deemed acceptable, which is unexpected if the anti-uniqueness condition (termed as *non-redundancy* in Blumberg (2020)) is a presupposition. Hence (12), contrasted with (13), suggests that the anti-uniqueness condition is not a presupposition.

In the next section, I turn to economy based approaches and discuss the challenges they encounter.

2.2. Accounts based on pragmatic economy principles

Blumberg (2020) and Ahn (2022, 2023) argue against encoding anti-uniqueness presuppositions in the denotation of demonstratives and instead posit that the markedness of demonstratives is a pragmatic effect arising out of reasons of economy. The core idea is that demonstratives are in competition with definite descriptions: a demonstrative may not be used if a definite can be. Blumberg contends that demonstratives take two arguments—the first argument is covert and the second argument is overt (i.e., the NP complement). The function of the covert argument is to pick out a particular individual from the set of individuals that satisfy the demonstrative's overt argument. Blumberg posits that this covert argument cannot be redundant (i.e., non-restricting). Put another way, the overt argument cannot already denote a singleton in context. Blumberg builds on Schlenker's (2005) *Minimize Restrictors!* to posit a competition principle that applies to the use of demonstratives. *Minimize Restrictors!* dictates that an expression such as *'the tall president'* is degraded when there is a unique president in the context, as this expression contains redundant material, ('*tall*'), that could be dropped without affecting the denotation of the expression. Hence it is blocked by the simpler description, *'the president'*. Similarly, a demonstrative with a non-restricting argument is argued to be ruled out because a simpler expression, the definite description, is available. Availability of both definites and demonstratives in cases such as (14) is explained by appealing to the notion of domain shifts. Whereas '*the*' selects a domain where there is a one car (or one girl), '*that*' selects a domain with multiple cars (or girls), and the covert argument of the demonstrative picks out a particular car (or girl) from that non-singleton set. This is very similar to Dayal and Jiang (2022)'s notion of domain widening, except Blumberg doesn't encode this in the meaning of demonstratives, but relies on linguistic context to bring about this spontaneous shift in domain.

(14) a. The/ That car is still parked outside our apartment. (Blumberg, 2020: 56)
b. Every boy who danced with a girl kissed the/ that girl. (Blumberg, 2020: 57)

Ahn's (2022; 2023) account draws on Blumberg (2020) but offers a different implementation. Ahn treats demonstratives as indirectly referential expressions with a binary maximality operator that takes two arguments, the NP complement and a deictic linker R, where R can be deictic pointing, or linguistically rendered deixis, such as an anaphoric index or a relative clause (15). The only semantic distinction between the definite and demonstrative is argued to be this linker argument. Definites lack this argument and use a unary supremum operator with a presuppositional index – it takes the NP restriction and returns the unique entity that meets that restriction (16). Thus the demonstrative is semantically more complex than the definite and, consequently, is dispreferred to definites when they resolve to the same meaning.

(15) $[\![\text{DEM}]\!] = [\![bi-sup]\!] = \lambda P \lambda R_\gamma \lambda w.\ \iota x : \forall y[P(y)(w) \wedge R_\gamma(y)(w_0) \leftrightarrow y \sqsubseteq x]$

(16) $[\![\text{DEF}]\!] = \lambda w \lambda P.\ \iota x : \forall y[P(x)(w) \leftrightarrow y \sqsubseteq x]$ (Ahn, 2023)

For instance, Ahn claims that the anti-uniqueness effects only arise for sentences like (4a) (repeated below in 17), in which the R slot for the demonstrative does not carry an overt element. In such cases, the definite and the demonstrative descriptions are extensionally equivalent, returning the unique author of Waverley. However, the demonstrative is semantically more complex in involving an additional argument, R, which is non-restricting and redundant. Thus, it is degraded due to pragmatic economy principles, as detailed above.

(17) The/#That author of Waverley also wrote Ivanhoe. (Nowak, 2019: 2867)

In the cases where the R slot is filled with contentful information like deixis, as in (12), deixis is at-issue and interpreted noun-internally for demonstratives. This contrast with definites, where if accompanying deixis is present, it plays a non-at-issue, supplementary role. Thus definites and demonstratives do not result in the same meaning in such cases, and, consequently, do not compete with each other. This accounts for the felicity of demonstrative descriptions in sentences like (12).

2.2.1. Limitations of pragmatic economy-based accounts

First, this account overgenerates in certain cases, which I present below. The account predicts that the inference of anti-uniqueness only arises when a definite description competes with the demonstrative, and hence we would not see anti-uniqueness effects in deictic (or generic) contexts. This is because they do not enter into a competition in such cases, since deixis is at-issue for demonstratives and not at-issue for definites. This prediction is claimed to be borne out in Ahn (2023). However, this does not seem to hold empirically. English speakers judge demonstratives in a generic context like (18) and in a deictic case like (19) to be infelicitous.

(18) The/#That sun rises in the east.

(19) [The/#That moon]$_{\rightarrow}$ very bright.

Second, Ahn (2022) claims that the demonstrative description *that author of Waverley* in (17) is felicitous when used anaphorically, as in (20). Ahn argues that the contrast between (17) and (20) is predicted on this account from the requirement that the demonstrative needs the R slot to be filled by at least one of three reference arguments: overt deixis, a relative clause, or an anaphoric index. When neither deixis nor a relative clause is present, demonstratives must at least carry an anaphoric index and are degraded in uniqueness contexts, where there is no anaphoric antecedent. In other words, the infelicity of '*that author of Waverley*' in (17) is because it is "an anaphoric demonstrative appearing in a non-anaphoric context" (Ahn, 2022: 1374), and the infelicity disappears when an anaphoric context is provided, as in (20).

(20) I once met the author of Waverley. That author of Waverley also wrote Ivanhoe. (Ahn, 2022: 1352)

I do not share Ahn's judgment for (20) and find the use of the demonstrative to be ruled out in this context, as shown in (21a). Crucially, a minimal change, as in (21b), can render the demonstrative felicitous.[6]

(21) a. I once met the author of Waverley. #That author of Waverley also wrote Ivanhoe.
b. I once met the author of Waverley. That author also wrote Ivanhoe.

Certainly, even the felicitous use of a demonstrative in (21b) is less preferred and comparatively degraded in comparison to other anaphoric options (such as a pronoun or a definite description; see Ahn (2019) for details), but in so far as the two demonstrative uses in (21a) and (21b) are compared, the contrast in felicity between them is stark. Notably, the relative acceptability of the demonstrative in (21b) (compared to the definite) improves markedly if we add *same*: '*I once met the author of Waverley. That* ***same*** *author also wrote Ivanhoe.* But, crucially, addition of *same* does not change the infelicity of the demonstrative in (21a). On the anti-uniqueness based approaches one could argue along the following lines to account for this data pattern: *that author* is felicitous because it allows potential contrast with other authors, since the set denoted by the predicate *author* is not inherently a singleton set (i.e., anti-uniqueness is satisfied). In

[6]Given the subtle nature of contrast in this domain, judgments for both (i) the infelicity of the demonstrative description in (21a) and (ii) the felicity of the demonstrative description in (21b) were sought from 7 other English speakers (5 linguists and 2 non-linguists), who corroborated the judgments reported here for (21).

comparison, *that author of Waverley* implies a contrast with *other authors of Waverley*, which cannot be accommodated given that the first sentence in (21a) uses a definite description to introduce the unique author of Waverley.[7] This would be in line with previous accounts of demonstratives that suggest that the use of a demonstrative implies that the property attributed to the referent does not hold for other members of the relevant contrast set (Roberts, 2002).

Lastly, I would like to highlight that the infelicity of the demonstrative in (21a) is significantly different from the relative markedness of the demonstrative in comparison to the other anaphoric options in a sentence like (22).

(22) A woman came into the room. She/The woman/That woman sat down.

While a pragmatic competition based account like Ahn (2019) (and subsequent works) can account for the relative preference between the three referential forms in (22), to explain the infelicity of demonstratives in (21a) we need something stronger than competition. If it was merely competition that was driving anti-uniqueness effects, then we would expect the behavior of demonstratives in anti-uniqueness cases like (21a) to mirror their behavior in anaphora cases like (22). In anaphoric contexts like (22), while definites are uncontroversially preferred over demonstratives, which are comparatively degraded, they (i.e., demonstratives) are certainly not ruled out, as shown in recent cross-linguistic experimental studies (Saha et al., 2023; Saha and Davidson, 2024). In comparison, demonstratives in anti-uniqueness violation contexts are judged to be patently infelicitous. This suggests that these two cases are distinct and calls for an alternate treatment of these constraints that distinguishes the stronger anti-uniqueness restriction of demonstratives from their general markedness in instances of felicitous use.

To summarize, in this section I discussed the two broad classes of approaches that account for the anti-uniqueness inference of demonstratives. I illustrated that both approaches have their own limitations. In the next section, I present evidence in support of the fundamental intuition behind anti-uniqueness theories.

3. Revisiting anti-uniqueness

3.1. Motivations

While we saw that the implementation of the anti-uniqueness condition in the existing accounts exhibit certain limitations, I argue that the anti-uniqueness theories capture the right intuition that should be retained in the foundations of any alternate proposals. To do so, let us take a step back. It was stated earlier that crucial to understanding the precise way in which definites and demonstratives differ is identifying specific contexts that tease apart their distribution. The discussion so far has focused on contexts where the definite description is licensed but the demonstrative is not. Let us now turn to the inverse cases, where the demonstrative is in its most natural habitat, and the definite is ruled out instead. To begin, let us consider the general role demonstratives commonly play in our everyday communication. Consider this excerpt from Nowak (2019), which shows that in many familiar cases, demonstratives supply answers to the question ‘which one?’ (p. 2872)

[7] I’m aware that there are several compositional barriers to a formal realization of this. This idea is not meant to serve as a theoretical foundation, but simply an intuitive starting point.

(23) A: Number 49? (Nowak, 2019: 2872)
B: Yes, I'd like a rib-eye steak, please.
A: Which one?
B: That one, please. (pointing)

This dialogue is an illustration of a typical use case of a deictic demonstrative description. As Nowak puts it, "by *saying 'rib-eye steak', B calls attention to a particular class of individuals, and by pointing, selects one from among them. Very frequently, when we use complex demonstratives deictically, a set of candidate referents is provided by the predicate from which the demonstrative is formed, and we pick out one by way of a gesture, or by its salience, or whatever*" (Nowak, 2019: 2872). This longstanding idea has been prominent in the philosophical literature as well (Reimer, 1991).

In anaphoric contexts, too, demonstratives are most natural, and in fact necessary, when overt contrast is established between linguistic antecedents that instantiate the same NP property (e.g., '*a quilt*' vs '*another quilt*' in (24a), '*a woman*' vs '*another woman*' in (24b)). The definite is naturally ruled out in both (23) and (24) as its uniqueness requirement cannot be met.

(24) a. I saw one quilt which was quite abstract, with lots of asymmetric diagonals. Another one was was more traditional, worked in an old Amish pattern. This quilt was less busy than the other, but just as bold. (Roberts, 2002: 5)
b. A woman entered from stage left. Another woman entered from stage right. That/ #the woman was carrying a basket of flowers. (Ionin et al., 2012: 73)

Now, let us look at slightly more complex cases, where there is no explicit contrast between linguistic antecedents in the discourse, yet, judgments seem to be robust that the demonstrative is felicitous, whereas the definite is unquestionably degraded in anaphoric reference.[8] This is illustrated in Srinivas (2024) with evidence from anaphoric uses of definites and demonstratives in kind-denoting contexts. For instance in (25a), the anaphoric use of '*the mouse*' is interpreted as not referring to the already mentioned class of wood mouse (but instead to the entire mouse-kind), and hence cannot be used anaphorically in this context. This contrasts with the use of the demonstratives, '*this mouse*' in (25a), and '*this (type of) butterfly*' in (25b), which exhibit no difficulty in referring back to the previously mentioned class of wood-mice and the previously mentioned class of monarch-butterfly respectively and are perfectly felicitous as anaphors.

(25) a. The wood mouse is a rodent native to Europe. #The/This mouse is a highly resourceful creature. (Srinivas 2024: 1)
b. The monarch butterfly is quite a remarkable creature. #The/ *The kind of/ This/ This kind of butterfly feeds on the nectar from many plants. (Srinivas 2024: 4)

This contrast is quite robust and extends to non-natural kinds, as in (26), and also to contexts involving an intended bridging relationship between the antecedent sentence and the subsequent definite or demonstrative phrase, as in (27) (Reed, 2024).

[8] I am grateful to Daria Bikina for bringing the Srinivas (2024) paper to my attention.

(26) Do you know [the kind of person that sings in the shower]$_k$? (Srinivas 2024: 5)
  a. I live with #the (*kind of) person$_k$.
  b. I live with that (kind of) person$_k$.

(27) (passion) is the most important quality in the making of a Major leaguer.
  a. Kevin Youkilis is that/ #the type of guy.
  b. Aaron Hill is that/ #the type of guy. (ex. 20a in Reed (2024))

The crux of Srinivas's (2024) proposal is the following: Srinivas argues that the infelicity of definite descriptions in contexts like (25a) and (25b) is due to a taxonomic domain restriction in the interpretation of kind-denoting uses of '*the*' (following Dayal (2004)). The relevant taxonomic domains for (25a) and (25b) are shown in Fig 1. The expressions '*the mouse*' and '*the butterfly*' fail to identify a unique kind inside the relevant taxonomic domain, thus their resulting infelicity as anaphors. (The proposal assumes a uniqueness based semantics for '*the*' in its anaphoric use).

Figure 1: LEFT: Taxonomic domain for (25a); RIGHT: Taxonomic domain for (25b) (Srinivas, 2024)

Crucially, for our purposes, we see that the sibling nodes at any level within the taxonomic hierarchy constitute a contrast set of different kinds of mice/butterfly. Similarly in (26), the context seem to support a natural partition of the set of individuals denoted by '*person*' into two kinds—{the kind that sings in the shower, the kind that does not sing in the shower}—thus providing a contrast set. The unifying factor for the perfectly felicitous examples of demonstrative, such as (23), (24), and (25), is that in all these cases a contrast set is available, whether by way of the presence of actual alternative referents the speaker could have pointed to (23), or via explicitly mentioned contrastive linguistic antecedents (24), or implicitly provided by the linguistic context (25). Thus the felicitous use of the demonstrative description in these contexts is anticipated on (some version of) the anti-uniqueness approaches, as the anti-uniqueness requirement can be met either in the extra-linguistic situation (23) or in the linguistic discourse—explicitly (24) or implicitly (25).

Having argued that there is sufficient evidence in language to support the core spirit of anti-uniqueness approaches, I now propose an implementation that can account for the challenges discussed in Section 2.1.1.

3.2. A modal implementation of anti-uniqueness

Recall that the anti-uniqueness presupposition as formalized in the existing literature fails to account for felicitous uses demonstratives in sentences like (11a) and (11b) (repeated below in (28a) and (28b)).

(28) a. That rhino is the only one still alive. (Nowak, 2019: 2897)
b. That atomic bomb is the last one remaining. (Nowak, 2019: 2897)

On a classical Kratzerian situation semantic view, these sentences result in presupposition failures on Dayal and Jiang's (2022) formulation (repeated below (29)): the largest possible situation is the world of evaluation (the maximal element). And the sentences assert that there is only one rhino or one bomb left in the world. Hence they should be infelicitous, contrary to speaker judgments.

(29) $[\![\text{DEM}]\!] = \lambda s \lambda P : \exists s' s \leq s' |P_{s'}| > 1. \iota x[P_s(x) \land x = y]$ (Dayal & Jiang, 2022: 158)

I argue that the use of the demonstrative in these cases actually implies that there is an *accessible alternate world* (not a larger situation) where the demonstrative's requirement for contrast (i.e., anti-uniqueness) is satisfied. And indeed, given the facts and history of our actual world, there is an accessible, conceivable counterfactual world where there is more than one rhino or more than one atomic bomb. In other words, I claim that the anti-uniqueness requirement of demonstratives is modal in nature.[9]

I will present further evidence for the modal nature of this constraint. Recall the felicity of the demonstrative in (12) (repeated below in (30a)). Blumberg (2020) and Ahn (2023) claim that the anti-uniqueness presupposition undergenerates as it predicts this sentence to be infelicitous, contrary to speaker judgment. Now consider the minimal pair in (30b), which is infelicitous.

(30) a. I don't know if there are any other cars in this showroom, but [that car]$_{\rightarrow}$ looks expensive. (Blumberg, 2020: 49)
b. #I **know** there aren't any other cars in this showroom, but [that car]$_{\rightarrow}$ looks expensive.

A modal approach can explain this contrast. As Blumberg states, in (30a) the speaker expresses doubt about the presence of cars in the showroom. Crucially, $doubt(p)$ does not rule out the existence of accessible possible worlds where p holds. If the possibility of p worlds is ruled out, it would essentially mean $know(\neg p)$, i.e., the case in (30b). This means that in (30a), there exists *some* possible world *w'* accessible from the world of context w_C (i.e., compatible with the speaker's knowledge in w_C), where there is more than one car in the showroom (in addition to there being accessible worlds where there is only one car in the showroom). In contrast, as we just established, in (30b), there is no possible world *w'* compatible with the speaker's knowledge in w_C, where there is more than one car in the showroom. That is, just as long as it is not ruled out that there exists an accessible world where the contrast requirement of the demonstrative is met, anti-uniqueness is satisfied.

[9]A possible way to rescue the situation semantic approach could be to posit that the anti-uniqueness presuppositions of (28a) and (28b) can be satisfied in some past situation. This could be potentially implemented by introducing a temporal trace function τ (Link, 1987), which takes an eventuality e (where e can be a situation in certain frameworks) and returns its run time. Hence, one could claim that there is an e such that $\tau(e) <_t$ 'now', when there were more than one rhino (and more than one atomic bomb), which satisfies the anti-uniqueness presupposition. However, I do not pursue this approach, as it does not account for a wider range of cases, where temporality is not the issue.

This line of reasoning is very similar to Blumberg (2020). Blumberg makes reference to possible worlds in his pragmatic account, and states that on the presuppositional approach the felicitous use of a demonstrative '*that F*' requires that every world in the context set be one in which F does not denote a singleton, which is too strict. In contrast, on the pragmatic approach using '*that F*' drives the weaker inference that it is not commonly believed that F denotes a singleton, otherwise the speaker would have used the '*the F*'.

As illustrated above, I agree with Blumberg that the anti-uniqueness condition is a weak one involving existential rather than universal quantification over accessible possible worlds; however I encode it as a presupposition. The motivation for this comes from the discussion in Section 2.2.1 that anti-uniqueness violations lead to very strong infelicity which is distinct from the general markedness of demonstratives in cases where definites and demonstratives overlap in their use (e.g., anaphora).

As a consequence of the weakly modalized presupposition proposed here, the problem that Blumberg (2020) raised for the existing presuppositional accounts of demonstratives does not hold. Recall that Blumberg argued that while (30a) violates the anti-uniqueness requirement of demonstratives, it is not infelicitous, suggesting that the violation is not a presupposition violation. This was contrasted with a clear case of presupposition violation, such as (13), (repeated below in (31)).

(31) #I don't know if Mary ever smoked, but Mary stopped smoking. (Blumberg, 2020: 49)

On the current explanation, the felicity of (30a) is readily explained by the fact that it does not violate a presupposition at all. As we just established, the presupposition of the demonstrative is satisfied in (30a), hence the felicity. In contrast, in (30b), there is a presupposition failure, and hence it is infelicitous, just like Blumberg's presupposition violation example in (31).

Note that a situation-semantic approach would overgenerate for (30b). It is certainly possible to conceive of a larger situation, extending beyond the car showroom, where the anti-uniqueness presupposition of the demonstrative could be met, and consequently (30b) would be predicted to be felicitous. As we see from the infelicity of (30b), this prediction is not borne out. This provides further evidence that a modal treatment is indeed the solution.

As with any modal phenomena, when we discuss the accessibility of possible worlds, the notion of ordering of worlds is important (Kratzer, 1981; von Fintel and Heim, 2007). For instance, it would be natural to assume that the worlds closest to our actual world would be the ones where states of affairs conform to our understanding of the norms and the natural states of our actual world: only one person can be the President of a country, our solar system has one sun, etc. Accessing possible worlds with multiple suns, or Presidents, requires willingness to be extremely permissive with our understanding of the norms and natural states of the world, which speakers are generally resistant to. But certainly these can be accommodated, given the right contextual support. Envision that you are Luke Skywalker, and you are in your home planet Tatooine, which has two suns, Tatoo I and Tatoo II (context from Owusu (2022)). Then it becomes certainly easy to imagine that you can felicitously use the demonstrative while pointing at Tatoo I, as in (32), even if Tatoo II is not visible in the sky in that context.

(32) [That sun]$_{\rightarrow}$ is bright.

Hence, I argue that the anti-uniqueness requirement needs to be re-framed in a possible worlds framework couched in terms of modal accessibility. This is implemented in the proposed denotation of the demonstrative in (33).

(33) $$[\![\text{DEM}]\!] = \lambda w \lambda P : \exists w' \in W[Acc(w)(w') = 1 \wedge |P_{w'}| > 1]. \begin{cases} \iota x[P_{w=w_C}(x) \wedge x = y_{n,w=w_C}] & (\mathbf{D}) \\ \iota x[P_w(x) \wedge \ x = y_i] & (\mathbf{A}) \end{cases}$$

I propose a bipartite denotation for demonstratives that separates their deictic (**D**) and anaphoric (**A**) uses. I assume that the two kinds of indices, exophoric ones that track pointing to entities outside the linguistic discourse and endophoric ones that track anaphoric links with linguistic antecedents, are conceptually distinct and hence are encoded into meaning differently. While demonstratives support both kinds of reference, their exophoric and anaphoric behavior is markedly different. In deictic contexts, demonstratives employ rigid exophoric indices to refer directly to entities in the external world, which are not sensitive to modal displacement. In contrast, anaphoric indices show flexibility in the entity it refers to across different scopal and modal relations. Note that the anti-uniqueness constraint holds for anaphoric uses as well, as in (34).

(34) Earth has a sun and a moon. #That sun is farther away from that moon.

In (33), the presupposition of the demonstrative is satisfied just as long as there is an accessible possible world from the world of context w_C, where a contrast set for P is available. As in previous versions, $x = y_{n,w=w_C}$ establishes an identity relation to the entity being pointed to, where $y_{n,w=w_C}$ picks the entity indicated by n^{th} act of demonstration by the speaker in w,[10] and w is fixed to w_C. Anchoring the world parameter to the world of the context w_C captures the fact that deictic demonstratives refer rigidly in w_C and is not subject to modal displacements.

While the parallels illustrated between the deictic and anaphoric uses of demonstratives in Section 3.1 strongly suggests that the core foundation of the meaning of demonstratives lie in contrast, and it is intuitively appealing to propose a unified semantics for their deictic and anaphoric uses, it is not a trivial task to implement this formally. In fact, there may be some merit in retaining a degree of separation between these two use cases. As discussed above, the deictic and anaphoric uses of referential expressions have some fundamental differences. Contrast in a deictic context can be understood as the 'presence of alternate entities in the world (or possible worlds) that could be pointed to' (i.e., a re-framing of anti-uniqueness). However, its parallel in anaphoric contexts ('presence of alternate linguistic antecedents that could be picked') does not have a well-defined meaning in cases where explicit contrast sets (e.g., '*this quilt*' vs '*another quilt*') are not linguistically provided. Nevertheless, we saw robust data in Section 3.1, which show that even in cases where there is no linguistically provided contrast set, demonstratives seem to exploit some notion of implicit contrast and are most favored in contexts that can support implicit contrast sets. Recall the kind-level anaphoric contexts, which readily accept demonstrative anaphora (while ruling out definite anaphora).

[10] Note that Ahn's (2022) account treats pointing as a location modifier, which restricts the location of the argument in the utterance context, instead of directly referring to an entity. I have followed Dayal and Jiang (2022) in retaining a directly referential treatment. However, I remain agnostic about the precise treatment of demonstration, as I believe it is orthogonal to the primary points of my claim here.

Furthermore, I would like to elaborate on a point raised in Section 2.2.1. Whereas violation of the anti-uniqueness condition of demonstratives results in downright infelicity, the markedness of demonstratives in anaphoric contexts show robust gradient effects. This has been shown in several recent experimental studies, which investigated the behavior of anaphoric demonstratives and definites cross-linguistically through controlled discourse manipulations (Saha, 2023; Saha et al., 2023; Saha and Davidson, 2024). For instance, one discourse factor that was shown to affect the acceptability of demonstratives is the presence of a contrasting noun in the context sentence, as in (35) (Saha et al., 2023: 467; adapted from Mandarin examples originally discussed in Jenks (2018) and Dayal and Jiang (2022)).

(35) a. **A boy** entered the classroom. The/ That boy sat down in the front row.
b. **A boy and a girl** entered the classroom. The/ That boy sat down in the front row.

In addition to, unsurprisingly, finding demonstratives to be relatively marked compared to definites in (35) (‘*The boy sat down...*’ rated significantly higher than ‘*That boy sat down...*’), they also found relative markedness between the use of demonstratives in (35a) vs (35b): demonstratives were rated significantly higher in (35a) than in (35b). Without going into the details of Saha et al.’s analysis, the key point to note is that a sentence like ‘*A boy and a girl entered the classroom*’ is the type of sentence that lends itself to a contrast set of the kind {boy, girl}, e.g., ‘*... The boy sat down, but the girl did not*’. On the other hand, as we saw in Section 3.1, demonstratives are most natural in contexts that supported contrast sets of the kind {that boy, another boy}. (See Saha et al. (2023) for more discussion.)

Comparison of the behavior of demonstratives in anti-uniqueness cases to the anaphora cases presented above suggests that though contrast incontrovertibly plays a role in the meaning of demonstratives, the manner in which contrast molds the semantics and pragmatics of demonstratives in these two cases is not identical. The overarching goal is to have an account that can capture the right division of labor between semantics and pragmatics in these uses of demonstratives. These investigations and the proposal for a comprehensive account that addresses the challenges raised here is currently being pursued in Saha (prep).

In the next section, I turn to a cross-linguistic investigation of anti-uniqueness. Generally, in existing literature anti-uniqueness has been taken to be a property of deictic elements like demonstratives (but see Owusu (2022) for Akan). I add additional complexity to the empirical landscape by illustrating that, in Bangla, anti-uniqueness is not just exhibited by demonstratives but also by bare classifier definites, which otherwise pattern with standard definites in not allowing deictic contrast. The data reported comes from the author, a native speaker.

4. Enriching the empirical landscape: definiteness in Bangla

Bangla is an Indo-Aryan classifier language, which expresses definiteness predominantly via the combination of a bare noun and a classifier [N-CL]. Indefiniteness is expressed when a noun composes with the numeral ‘one’ via the interpolation of a classifier, as shown in (36).

(36) ek-ta baccha
one-CLF child
‘One child’/ ‘A child’

In definite uses, the numeral is dropped and the NP raises to the specifier of a +def DP, giving rise to the [N-CL] construction (Bhattacharya, 1999a, b; Dayal, 2012, 2014). These bare classifier constructions (term due to Simpson (2005)) are used to in both uniqueness and anaphoric contexts in the language, as shown in (37a) and (37b).

(37) a. gram-er mudir dokan-ta bondho.
village-GEN grocery store-CL close
'The grocery store in the village is closed.' (uniqueness)

b. ek-ta chele esheshhe. chele-ta mishuke.
one-CL boy has-come boy-CL friendly
'A boy has come. The boy is friendly.' (anaphoricity)

Unlike English definites and similar to English demonstratives, bare classifier definites exhibit anti-uniqueness and are not felicitous for globally unique nouns (38a). Demonstrative descriptions in Bangla, which are of the form [DEM N-CL], are also ruled out in such cases (38b). Only bare nouns are felicitous in these cases, as in (38c). (See Yip et al. (2023) for more discussion.)

(38) a. #surjo-ta pub dik-e othe.
sun-CL east side-LOC rise.HAB
Intended: 'The sun rises in the east.'

b. #oi surjo-ta pub dik-e othe.
That sun-CL east side-LOC rise.HAB
Intended: 'That sun rises in the east.'

c. surjo pub dik-e othe.
sun east side-LOC rise.HAB
'The sun rises in the east.'

Bare classifier definites exhibit anti-uniqueness effects, like demonstrative do with inherently unique nouns. However, unlike English demonstratives (39a), and like English definites (39b), they do not allow deictic contrast (39c).

(39) a. I like [that car]$_{\rightarrow}$, but I don't like [that car]$_{\rightarrow}$

b. #I like [the car]$_{\rightarrow}$, but I don't like [the car]$_{\rightarrow}$.

c. #jokhon kukur-ta$_{\rightarrow}$ ghumochilo, tokhon kukur-ta$_{\rightarrow}$ dourochilo
when dog-CLF.SG sleep.PST.PROG.3 then dog-CLF.SG run.PST.PROG.3
Intended: 'When the dog I am pointing at was sleeping, the dog I am pointing at was running around.'

To recapitulate, Bangla bare classifier definites pattern with standard demonstratives vis-à-vis anti-uniqueness, but they align with standard definites in not allowing contrastive deixis. Interestingly Bangla is not unique in this respect. We see strikingly parallel cross-linguistic evidence for this divergent empirical pattern in another unrelated language, Akan, where the determiner *nó* exhibits anti-uniqueness effects, as shown in (40a), but does not allow contrastive pointing, as shown in (40b). (See Owusu (2022) for a detailed discussion.)

(40) a. Awia (#nó) yɛ nsoroma. (Owusu, 2022: 18)
sun DEF COP star
'The sun is a star.'

b. #Me-pɛ car nó nanso me-m-pɛ car nó (Bombi, 2018: 152)
1SG-like car DEF but 1SG-NEG-like car DEF
Intended: 'I like that car [pointing at Audi] but I don't like that car [pointing at Renault].'

This pattern of data calls for the dissociation of the anti-uniqueness property from the deictic component of referential linguistic expressions. Consequently, it poses a challenge for the analysis in Ahn (2022, 2023). On Ahn's analysis, the presence of the additional deictic linker R, which is present in demonstratives and absent in definites, is the crucial element in deriving the anti-uniqueness inferences of demonstratives. This is also the element that hosts deictic pointing. So Ahn's analysis predicts that anti-uniqueness effects would only be exhibited by linguistic elements that also require deixis (and hence allow for deictic contrast). As the data from Bangla and Akan show, this prediction is not borne out cross-linguistically.

On the other hand, anti-uniqueness approaches are better suited to capture such data, since the anti-uniqueness condition is encoded as a separate restriction. Hence, adapting the proposed semantics for demonstratives in (33), the denotation of the Bangla bare classifier definite is proposed as follows:

(41) $$[\![\text{N-CL}]\!] = \lambda w \lambda P : \exists w' \in W[Acc(w)(w') = 1 \wedge |P_{w'}| > 1].\begin{cases} \iota x[P_w(x)] & \textit{uniqueness} \\ \iota x[P_w(x) \wedge\ x = y_i] & \textit{anaphoricity} \end{cases}$$

The proposed denotation of N-CL definites in (41) captures their similarity to demonstratives in the case of inherently unique nouns by encoding an anti-uniqueness presupposition in their denotation. At the same time, unlike demonstratives, they do not encode (exophoric) deictic indices, thus cannot establish a deictic identity relation ($x = y_{n,w}$) established by pointing – this explains their incompatibility with deictic contrast. They, however, do encode endophoric indices, which allow for their anaphoric use.

5. Conclusion

In conclusion, I have illustrated that a pragmatic economy based account fails to distinguish the stronger anti-uniqueness inference from the general markedness of demonstratives and that a presuppositional approach to anti-uniqueness is better suited to address this distinction. Crucially, I have proposed that the anti-uniqueness condition of demonstratives must necessarily be understood as a modalized restriction and have developed an implementation that tackles previous challenges. Lastly, providing cross-linguistic evidence, I established the need for decoupling the anti-uniqueness property of referential linguistic expressions from their deictic component and showed that a presuppositional treatment of anti-uniqueness can achieve this.

References

Ahn, D. (2017). Definite and demonstrative descriptions: a micro-typology. In *Proceedings of GLOW in Asia XI*, Volume 1, pp. 33–48. MIT Working Papers in Linguistic.

Ahn, D. (2019). *That thesis: A competition mechanism for anaphoric expressions*. Ph. D. thesis.

Ahn, D. (2022). Indirectly direct: An account of demonstratives and pointing. *Linguistics and Philosophy 45*(6), 1345–1393.

Ahn, D. (2023). Definite expressions with and without deixis. In *Proceedings of the West Coast Conference on Formal Linguistics (WCCFL)*, Volume 41, pp. 1–17.

Bach, E. and R. Cooper (1978). The np-s analysis of relative clauses and compositional semantics. *Linguistics and Philosophy 2*(1), 145–150.

Bhattacharya, T. (1999a). *The structure of the Bangla DP*. Ph. D. thesis, University College London.

Bhattacharya, T. (1999b). *The Structure of the DP in Bangla*. Ph. D. thesis, University of London.

Blumberg, K. H. (2020). Demonstratives, definite descriptions and non-redundancy. *Philosophical Studies 177*, 39–64.

Bombi, C. (2018). Definiteness in akan: Familiarity and uniqueness revisited. *Semantics and Linguistic Theory (SALT) 28*, 141–160.

Bremmers, D., J. Liu, M. van der Klis, and B. Le Bruyn (2022). Translation mining: Definiteness across languages (a reply to Jenks 2018). *Linguistic Inquiry 53*(4), 735–752.

Dayal, V. (2004). Number marking and (in) definiteness in kind terms. *Linguistics and Philosophy 27*, 393–450.

Dayal, V. (2012). Bangla classifiers: Mediating between kinds and objects. *Rivista di Linguistica 24*(2), 195–226.

Dayal, V. (2014). Bangla plural classifiers. *Language and Linguistics 15*(1), 47–87.

Dayal, V. (2023). Bare arguments cross-linguistically: Kind terms, definites, indefinites. 71st Linguistic Society of America (LSA) Summer Institute. Lecture handout.

Dayal, V. (in prep). *The Open Handbook of (In)definiteness: A Hitchhiker's Guide to Interpreting Bare Arguments*. Open Handbooks in Linguistics. Cambridge, MA: MIT Press.

Dayal, V. and L. J. Jiang (2022). The puzzle of anaphoric bare nouns in mandarin: A counterpoint to index! *Linguistic Inquiry 54*, 147–167.

Dryer, M. S. (2011). Definite articles. In M. S. Dryer and M. Haspelmath (Eds.), *The World Atlas of Language Structures Online*. Max Planck Digital Library. [Online].

Elbourne, P. (2008). Demonstratives as individual concepts. *Linguistics and Philosophy 31*, 409–466.

Himmelmann, N. P. (1996). Demonstratives in natural discourse: A taxonomy of universal uses. *Studies in anaphora 33*, 205.

Jenks, P. (2015). Two kinds of definites in numeral classifier languages. In *Semantics and Linguistic Theory*, Volume 25, pp. 103–124.

Jenks, P. (2018). Articulated definiteness without articles. *Linguistic Inquiry 49*(3), 501–536.

Kadmon, N. and F. Landman (1993). Any. *Linguistics and Philosophy*, 353–422.

Kaplan, D. (1989). Demonstratives: An essay on the semantics, logic, metaphysics and epistemology of demonstratives and other indexicals. In J. Almog, J. Perry, and H. Wettstein (Eds.), *Themes From Kaplan*, pp. 481–563. Oxford University Press.

King, J. (2001). *Complex Demonstratives*. Cambridge, MA: MIT Press.

Kratzer, A. (1981). The notional category of modality. In H.-J. Eikmeyer and H. Rieser (Eds.), *Words, Worlds, and Contexts: New Approaches in Word Semantics*, pp. 38–74. Berlin: De

Gruyter.
Kratzer, A. (1989). An investigation of the lumps of thought. *Linguistics and philosophy 12*(5), 607–653.
Lakoff, R. (1974). Remarks on this and that. In *Proceedings of the Chicago Linguistics Society*, Volume 10, pp. 345–356.
Link, G. (1987). Algebraic semantics of event structures. In J. Groenendijk, M. Stokhof, and F. Veltman (Eds.), *Proceedings of the Sixth Amsterdam Colloquium*, Amsterdam, pp. 243–262. University of Amsterdam.
Nichols, J. (1988). *An Ojibwe Text Anthology*. London, Ontario: Center for Research and Teaching of Canadian Native Languages.
Nowak, E. (2019). Complex demonstratives, hidden arguments, and presupposition. *Synthese 198*, 2865–2900.
Owusu, A. P. (2022). *Cross-Categorial definiteness/ Familiarity*. Ph. D. thesis, Rutgers The State University of New Jersey, School of Graduate Studies.
Partee, B. (1975). Montague grammar and transformational grammar. *Linguistic Inquiry 6*(2), 203–300.
Potts, C. and F. Schwarz (2010). Affective 'this'. *Linguistic Issues in Language Technology 3*(5), 1–30.
Reed, A. (2024). Procedural, underspecified, but good-enough, the. Manuscript.
Reimer, M. (1991). Demonstratives, demonstrations, and demonstrata. *Philosophical Studies 63*, 187–202.
Roberts, C. (2002). Demonstratives as definites. In K. van Deemter and R. Kibble (Eds.), *Information Sharing: Reference and Presupposition in Language Generation and Interpretation*, pp. 89–196. Stanford: CSLI Press.
Robinson, H. M. (2005). *Unexpected (in) definiteness: Plural generic expressions in Romance*. Ph. D. thesis, Rutgers The State University of New Jersey, School of Graduate Studies.
Ross, J. R. (1967). *Constraints on Variables in Syntax*. Ph.d. thesis, MIT.
Saha, A. (2023). The anaphoric potential of demonstrative descriptions: An experimental study. Generals paper, Harvard University.
Saha, A. (in prep). *The Anatomy of Definiteness*. Ph. D. thesis, Harvard University.
Saha, A. and K. Davidson (2024). Distinctions in anaphoricity: German strong definites vs. demonstratives. *Poster at the 5th Experimental Pragmatics in Italy Conference (XPRAG.it 2024)*.
Saha, A., Y. Sağ, J. Cui, and K. Davidson (2024). Anaphoric demonstratives in Mandarin. In *Proceedings of Semantics and Linguistic Theory (SALT)*, Volume 34, pp. 213–232.
Saha, A., Y. Sağ, and K. Davidson (2023). Focus on demonstratives: Experiments in English and Turkish. In *Semantics and Linguistic Theory (SALT)*, Volume 33, pp. 460–479.
Schlenker, P. (2005). Minimize restrictors! In *Proceedings of SuB9*, pp. 385.
Schwarz, F. (2009). *Two types of definites in natural language*. Ph. D. thesis.
Schwarz, F. (2013). Two kinds of definites cross-linguistically. *Language and Linguistics Compass 7*(10).
Simonenko, A. (2014). *Grammatical ingredients of definiteness*. Ph. D. thesis, McGill University.
Simpson, A. (2005). Classifiers and DP structure in Southeast Asia. In G. Cinque and R. S. Kayne (Eds.), *The Oxford Handbook of Comparative Syntax*, pp. 806–838. Oxford: Oxford

University Press.
Simpson, A. and Z. Wu (2022). Constraints on the representation of anaphoric definiteness in mandarin chinese. *New explorations in Chinese theoretical syntax: Studies in honor of Yen-Hui Audrey Li, Linguistik Aktuell/Linguistics Today 272*, 301–330.
Srinivas, S. (2024). The does not encode an anaphoric index: Evidence from kind uses. *Proceedings of the Linguistic Society of America 9*(1), 5764.
Stockwell, R. P., P. Schachter, and B. Partee (1973). *The Major Syntactic Structures of English*. New York: Holt, Rinehart and Winston.
von Fintel, K. and I. Heim (2007). Intensional semantics. *Lecture Notes*.
Wolter, L. (2006). *That's that: The semantics and pragmatics of demonstrative noun phrases*. Ph. D. thesis, University of California, Santa Cruz.
Yip, K.-F., U. Banerjee, and M. C. Y. Lee (2023). Are there "weak" definites in bare classifier languages? *Semantics and Linguistic Theory (SALT) 33*, 253–275.